Majid Farhadi

farhadi@ualberta.ca Edmonton, Alberta, Canada majidfarhadi78 majid-farhadi

Education

University of Alberta, Computer Engineering — Edmonton, Canada — 2023 – 2026

Isfahan University of Technology, Computer Engineering — Isfahan, Iran — 2018 – 2023

Experience

Very Large Scale Integration (VLSI) Research Lab, University of Alberta, Graduate Research Assistant — Edmonton, Canada — 2023 – 2026 — 3 years

- Designed and implemented AI architectures, including training and inference circuits, on FPGAs.
- Integrated and synchronized IP core modules with edge detector 1-stage synchronizers.
- Wrote testbench modules for behavioral, functional, and timing simulation and verification of RTL designs.
- Defined implementation constraints including clock period to meet timing requirements.
- Optimized dynamic power by minimizing switching activity through stop conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

Structured Machine Learning Research Lab, Simon Fraser University, Graduate Research Assistant — Burnaby, Canada — 2023 – 2024 — 1 year

- Counted motifs in a graph database by matrix multiplication, suitable for ML applications.